

[5] Leveraging Large Language Models to Generate Course-Specific Semantically Annotated Learning Objects

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1-Minute Executive Takeaway

Key Finding: Shows that automated CS question generation still needs human instructor oversight, justifying our asynchronous Moodle staging design.

Core Research Objective

To evaluate the feasibility and quality of LLM-generated computer science assessment objects grounded in university course materials.

Methodology & Architecture

Evaluated generated programming learning objects using computer science education faculty raters across multiple cognitive complexity levels.

Key Empirical Findings & Evidence

Found that while LLMs generate syntactically plausible items, approximately 15–25% of generated computer science questions still require manual human intervention due to subtle conceptual errors or misaligned distractors.

Direct Relevance to Our Kwiz Paper

Directly cited in Section 1, Section 2.2, and Section 3.4. Lohr et al.'s finding that expert intervention remains necessary directly motivated our asynchronous human-in-the-loop workflow: generated items are staged in Moodle for instructor review rather than published directly to active student quizzes.